



Written examination: Fall 2024

Course number: 02450

Course name: Introduction to Machine Learning and Data Mining

Aids allowed: No aids, except two A4 sheets with handwritten notes (front and back), standard dictionaries, and a non-programmable calculator.

Exam duration: 4 hours.

Format: The exam is multiple choice. All questions have four possible answers marked by the letters A, B, C, and D as well as the answer “Don’t know” marked by the letter E. A correct answer gives 3 points, a wrong answer gives -1 point, and “Don’t know” (E) gives 0 points.

Weighting: Exam questions are weighted equally.

Answers This exam only allows for physical hand-in. Select **one** answer per exam question by completely filling in a circle (e.g. ☐ ☒ ☐ ☐ ☐) using a black/dark pencil or pen. Do **not** use checkmarks or crosses to indicate your desired answer). To change an answer either erase the previously marked answer using an eraser, or clearly cross over the answer and fill another circle and write the letter of the final answer at the end of the line, e.g. ☐ ☒ ☐ ☒ ☐ D). Answer E if you do not wish to provide an answer for a question.

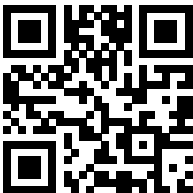
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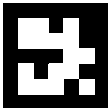
Student ID:

Legibly write you student ID (without the s) e.g. 123456 in the form and fill in the corresponding circle below the handwritten digit.

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	A	B	C	D	E
Question 1	☐	☐	☐	☐	☐
Question 2	☐	☐	☐	☐	☐
Question 3	☐	☐	☐	☐	☐
Question 4	☐	☐	☐	☐	☐
Question 5	☐	☐	☐	☐	☐
Question 6	☐	☐	☐	☐	☐
Question 7	☐	☐	☐	☐	☐
Question 8	☐	☐	☐	☐	☐
Question 9	☐	☐	☐	☐	☐
Question 10	☐	☐	☐	☐	☐
Question 11	☐	☐	☐	☐	☐
Question 12	☐	☐	☐	☐	☐
Question 13	☐	☐	☐	☐	☐
Question 14	☐	☐	☐	☐	☐
Question 15	☐	☐	☐	☐	☐
Question 16	☐	☐	☐	☐	☐
Question 17	☐	☐	☐	☐	☐
Question 18	☐	☐	☐	☐	☐
Question 19	☐	☐	☐	☐	☐
Question 20	☐	☐	☐	☐	☐
Question 21	☐	☐	☐	☐	☐
Question 22	☐	☐	☐	☐	☐
Question 23	☐	☐	☐	☐	☐
Question 24	☐	☐	☐	☐	☐
Question 25	☐	☐	☐	☐	☐
Question 26	☐	☐	☐	☐	☐
Question 27	☐	☐	☐	☐	☐



No.	Attribute description	Abbrev.
x_1	Magnitude in the range 2-6 (in ML)	Magnitude
x_2	Depth in the range 0-200 (in km)	Depth
x_3	Latitude in the range 43.7 to 48.2 (in Degrees)	Latitude
x_4	Longitude in the range 20.7 to 29.7 (in Degrees)	Longitude
y	The strength of the earthquake	Earthquake level

Table 1: Description of the attributes of the Earthquakes in Romania dataset used in this exam. The dataset contains all the earthquakes that originated in Romania since late 2012 until today. It includes the magnitude as measured by Richter local magnitude (ML), the depth of the earthquake, the latitude and longitude. The dataset used here consists of $N = 3714$ observations and the attribute y is discrete such that $y = 1$ (corresponds to Low strength), and $y = 2$ (corresponds to High strength).

Question 1. The main dataset used in this exam is the Earthquakes in Romania dataset¹ described in Table 1. We consider the four attributes of the Earthquakes in Romania dataset in Table 1. A scatter plot matrix for these attributes is given in Figure 1.

Which one of the following matrices represents the empirical correlation matrix for these attributes?

A. $\hat{C} \approx \begin{bmatrix} 1.0 & 0.48 & 0.98 & 0.21 \\ 0.48 & 1.0 & 0.02 & 0.46 \\ 0.98 & 0.02 & 1.0 & 0.07 \\ 0.21 & 0.46 & 0.07 & 1.0 \end{bmatrix}$

B. $\hat{C} \approx \begin{bmatrix} 1.0 & -0.48 & 0.02 & 0.21 \\ -0.48 & 1.0 & 0.02 & 0.46 \\ 0.02 & 0.02 & 1.0 & 0.07 \\ 0.21 & 0.46 & 0.07 & 1.0 \end{bmatrix}$

C. $\hat{C} \approx \begin{bmatrix} 1.0 & 0.48 & 0.02 & 0.21 \\ 0.48 & 1.0 & 0.02 & 0.46 \\ 0.02 & 0.02 & 1.0 & 0.07 \\ 0.21 & 0.46 & 0.07 & 1.0 \end{bmatrix}$

D. $\hat{C} \approx \begin{bmatrix} 1.0 & 0.02 & 0.02 & 0.21 \\ 0.02 & 1.0 & 0.02 & 0.46 \\ 0.02 & 0.02 & 1.0 & 0.07 \\ 0.21 & 0.46 & 0.07 & 1.0 \end{bmatrix}$

E. Don't know.

¹Dataset obtained from www.kaggle.com/datasets/stefancomanita/earthquakes-in-romania-from-2012-to-2024

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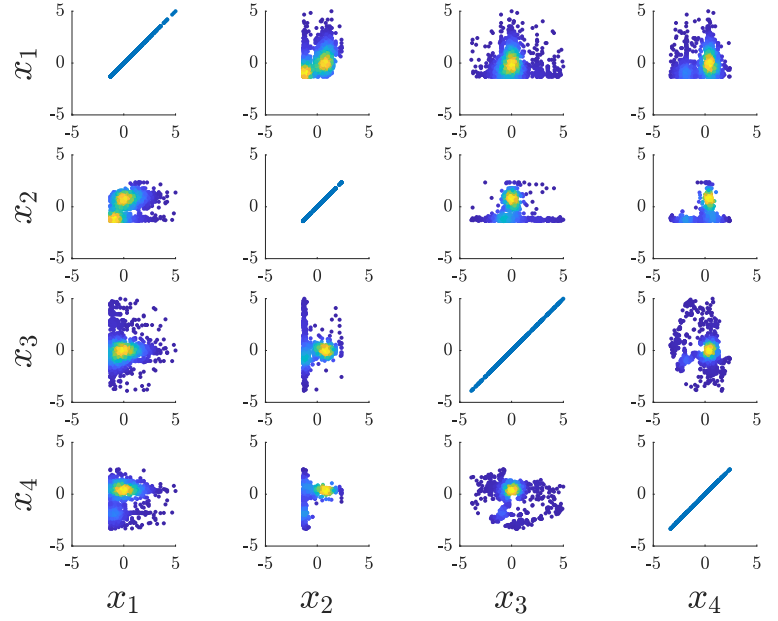


Figure 1: The scatter plot matrix for the attributes x_1 , x_2 , x_3 and x_4 of the Earthquakes in Romania dataset of Table 1. In the off-diagonal panels, the color represents the associated density.

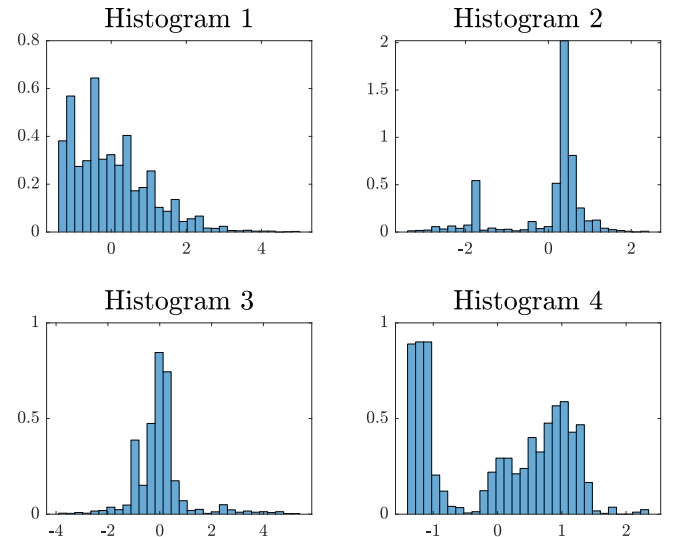


Figure 2: Histograms of the attributes in which their order has been permuted.

Question 2. We consider the histograms of the four attributes in Figure 2. Match the histograms to the attributes by using the information regarding the distributions of the points from Figure 1.

Which one of the following statements is *correct*?

- A. Histogram 1: x_3 , Histogram 2: x_4
Histogram 3: x_2 , Histogram 4: x_1 .
- B. Histogram 1: x_1 , Histogram 2: x_4
Histogram 3: x_3 , Histogram 4: x_2 .
- C. Histogram 1: x_2 , Histogram 2: x_1
Histogram 3: x_3 , Histogram 4: x_4 .
- D. Histogram 1: x_3 , Histogram 2: x_2
Histogram 3: x_4 , Histogram 4: x_1 .
- E. Don't know.

Question 3. Consider the PCA analysis for the Earthquakes in Romania dataset, in particular the SVD decomposition of the centered data matrix $\tilde{\mathbf{X}}$, and the corresponding matrix

$$\mathbf{V} = \begin{bmatrix} 0.54 & -0.13 & 0.68 & -0.48 \\ 0.65 & -0.08 & 0.0 & 0.76 \\ 0.07 & 0.98 & 0.16 & 0.04 \\ 0.53 & 0.09 & -0.72 & -0.44 \end{bmatrix}.$$

Which one of the following statements is *correct*?

- A. An observation with a high value of x_1 (**Magnitude**), a high value of x_2 (**Depth**), and a high value of x_4 (**Longitude**) will typically have a negative value of the projection onto principal component number 1.
- B. An observation with a low value of x_1 (**Magnitude**), and a high value of x_4 (**Longitude**) will typically have a negative value of the projection onto principal component number 3.
- C. An observation with a low value of x_3 (**Latitude**) will typically have a positive value of the projection onto principal component number 2.
- D. An observation with a low value of x_1 (**Magnitude**), a high value of x_2 (**Depth**), and a low value of x_4 (**Longitude**) will typically have a negative value of the projection onto principal component number 4.
- E. Don't know.

	x_1	x_2	x_3	x_4	y
$\mathbf{o}_1$	0	0	0	0	1
$\mathbf{o}_2$	0	0	1	1	1
$\mathbf{o}_3$	1	1	0	0	1
$\mathbf{o}_4$	0	1	1	1	1
$\mathbf{o}_5$	1	1	0	1	2
$\mathbf{o}_6$	0	0	0	0	2
$\mathbf{o}_7$	1	1	0	1	2
$\mathbf{o}_8$	0	0	0	0	2

Table 2: The observations $\mathbf{o}_i$ together with the binary attributes x_j and the associated label information y .

Question 4. We consider the dataset in Table 2 with the binary attributes x_j for $j = 1, \dots, 4$. We construct a Bayes classifier relying on the attributes x_2 and x_4 .

Which one of the following probabilities is *correct*?

- A. $p(y = 2 \mid x_2 = 1, x_4 = 1) \approx 0.67$
- B. $p(y = 2 \mid x_2 = 1, x_4 = 1) \approx 0.5$
- C. $p(y = 2 \mid x_2 = 1, x_4 = 1) \approx 0.39$
- D. $p(y = 2 \mid x_2 = 1, x_4 = 1) \approx 0.92$
- E. Don't know.

Question 5. Which one of the following statements about logistic regression is *correct*?

- A. The attributes of a logistic regression model cannot be modified in order to construct classifiers with nonlinear decision boundaries.
- B. In terms of a bias-variance trade-off, a logistic regression model with a well-tuned regularization parameter has a negligible bias but a fairly high variance.
- C. Consider a regularized logistic regression classifier. For a fixed value of λ , there exists a closed-form solution for the parameters $\mathbf{w}$, obtained by differentiating the objective function and setting the derivative to zero.
- D. Consider a regularized logistic regression model for a binary classification problem with a balanced dataset. If the estimated test error is approximately 0.5, this typically indicates over-regularization.
- E. Don't know.

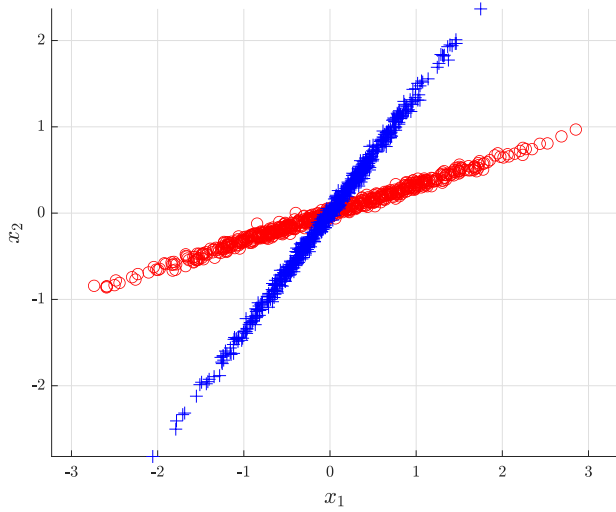


Figure 3: A two dimensional dataset of two classes indicated by blue (+) markers and red (o) markers.

Question 6. Suppose a PCA analysis is carried out on a two dimensional dataset presented in Figure 3 where two classes can be distinguished. In Figure 4 the projections of the dataset onto four different directions are presented, one per panel. Two of these plots correspond to the projection on the first and the second principal component.

Which one of the options below is *correct*?

- A. First principal direction corresponds to Plot 3
Second principal direction corresponds to Plot 4
- B. First principal direction corresponds to Plot 1
Second principal direction corresponds to Plot 4
- C. First principal direction corresponds to Plot 1
Second principal direction corresponds to Plot 2
- D. First principal direction corresponds to Plot 2
Second principal direction corresponds to Plot 1
- E. Don't know.

Question 7. We consider a binary classification problem where the input has two attributes $\mathbf{x} = [x_1, x_2]^\top$. We model the class conditional distributions with the following multivariate normal distributions

$$p(\mathbf{x} \mid y = j) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}^{(j)}, \boldsymbol{\Sigma}^{(j)})$$

$$\boldsymbol{\mu}^{(j)} = \begin{bmatrix} \mu_1^{(j)} \\ \mu_2^{(j)} \end{bmatrix}, \quad \boldsymbol{\Sigma}^{(j)} = \begin{bmatrix} \Sigma_{11}^{(j)} & \Sigma_{12}^{(j)} \\ \Sigma_{21}^{(j)} & \Sigma_{22}^{(j)} \end{bmatrix}, \quad \text{for } j = 1, 2.$$

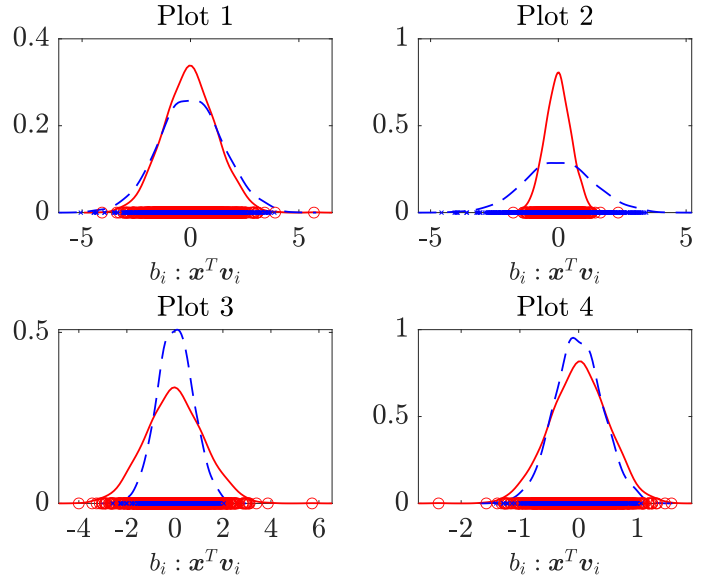


Figure 4: The projection of the dataset onto four different vectors. Note that each class is represented with a different density in each plot. In particular, the red class with marker (o) is represented by the continuous density curve, and the blue class with marker (+) is represented by the dashed density curve.

We use the *Naive Bayes* approach to derive a decision rule for classifying a point $\mathbf{x}$ in *Class 1*.

Which one of the following statements is *correct*?

Hint: Consider the ratio of the probabilities.

- A. $\frac{(x_1 - \mu_1^{(2)})^2}{\Sigma_{11}^{(2)}} + \frac{(x_2 - \mu_2^{(2)})^2}{\Sigma_{22}^{(2)}} - \frac{(x_1 - \mu_1^{(1)})^2}{\Sigma_{11}^{(1)}} - \frac{(x_2 - \mu_2^{(1)})^2}{\Sigma_{22}^{(1)}} + \log \Sigma_{11}^{(2)} + \log \Sigma_{22}^{(2)} - \log \Sigma_{11}^{(1)} - \log \Sigma_{22}^{(1)} > 2 \log \frac{p(y=2)}{p(y=1)}$
- B. $\frac{(x_1 - \mu_1^{(2)})^2}{\Sigma_{11}^{(2)}} + \frac{(x_2 - \mu_2^{(2)})^2}{\Sigma_{22}^{(2)}} - \frac{(x_1 - \mu_1^{(1)})^2}{\Sigma_{11}^{(1)}} - \frac{(x_2 - \mu_2^{(1)})^2}{\Sigma_{22}^{(1)}} + \log \Sigma_{11}^{(2)} + \log \Sigma_{22}^{(2)} - \log \Sigma_{11}^{(1)} - \log \Sigma_{22}^{(1)} > 2 \log \frac{p(y=1)}{p(y=2)}$
- C. $(\mathbf{x} - \boldsymbol{\mu}^{(2)})^\top (\boldsymbol{\Sigma}^{(2)})^{-1} (\mathbf{x} - \boldsymbol{\mu}^{(2)}) + \log |\boldsymbol{\Sigma}^{(2)}| - (\mathbf{x} - \boldsymbol{\mu}^{(1)})^\top (\boldsymbol{\Sigma}^{(1)})^{-1} (\mathbf{x} - \boldsymbol{\mu}^{(1)}) - \log |\boldsymbol{\Sigma}^{(1)}| > 2 \log \frac{p(y=2)}{p(y=1)}$
- D. $(\mathbf{x} - \boldsymbol{\mu}^{(2)})^\top (\boldsymbol{\Sigma}^{(2)})^{-1} (\mathbf{x} - \boldsymbol{\mu}^{(2)}) + \log |\boldsymbol{\Sigma}^{(2)}| - (\mathbf{x} - \boldsymbol{\mu}^{(1)})^\top (\boldsymbol{\Sigma}^{(1)})^{-1} (\mathbf{x} - \boldsymbol{\mu}^{(1)}) - \log |\boldsymbol{\Sigma}^{(1)}| > 2 \log \frac{p(y=1)}{p(y=2)}$
- E. Don't know.

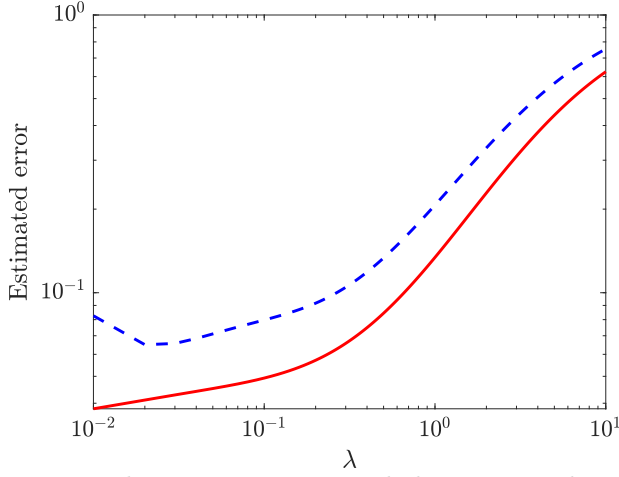


Figure 5: The training error and the estimated generalization error.

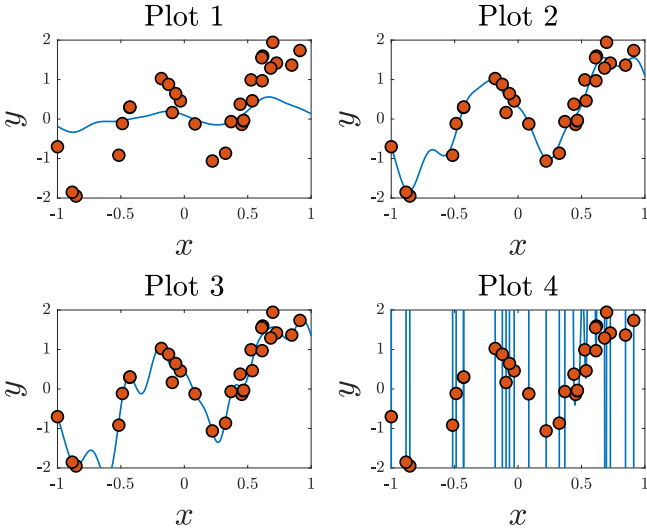


Figure 6: Each plot corresponds to the learned function $f(x; \mathbf{w})$ shown as a blue curve for a different λ -value.

Question 8. We are given a dataset with N observations for a regression problem and we fit a regularized linear regression model $f(x; \mathbf{w})$ with nonlinear transformation of the attributes and parameters $\mathbf{w}$. The cost function is written as

$$E(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N L(f(x_i; \mathbf{w}), y_i) + \lambda \Omega(\mathbf{w}),$$

where the loss $L(f(x; \mathbf{w}), y) \geq 0$ measures the distance between the true label y and the corresponding predicted value $f(x; \mathbf{w})$, and the regularizer $\Omega(\mathbf{w}) \geq 0$ penalizes monotonically the weights i.e., the higher the weights, the higher the $\Omega(\mathbf{w})$. We apply K -fold cross-validation to estimate the generalization error, and the learning curves are shown in Figure 5 with the corresponding learned models in Figure 6.

Which one of the statements below is *correct*?

- A. Considering Figure 6, Plot 1 corresponds to low values of λ and Plot 4 to high values of λ .
- B. In Figure 5 the dashed blue curve represents the train error and the continuous red the test error.
- C. Plot 3 corresponds to a solution with higher variance compared to Plot 2.
- D. When the cost function has the form as above, for any loss and regularizer, we can *always* compute the optimal weights as $\mathbf{w}^* = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \hat{\mathbf{y}}$.
- E. Don't know.

Question 9. Which one of the statements is *correct*?

- A. For least squares linear regression the test error will always decrease as we include more nonlinear transformation of attributes in the model.
- B. Consider a dataset with N observations. Leave-one-out cross-validation will require fitting N models in total to find the best value of $\lambda \in [0, 10]$.
- C. Regularized linear regression always results in a better test error than unregularized linear regression regardless of the λ -value.
- D. Apart from regularization we can apply a bagging style approach to reduce the variance and increase the bias of a linear regression model with nonlinear transformation on the attributes.
- E. Don't know.

Question 10. A regularized linear regression model is trained using the regularization term $\|\mathbf{w}\|_2^2$ with parameter $\lambda = 9$. We learn the parameters of the model based on the standardized training data $\hat{\mathbf{X}}$, where the mean and standard deviation of the training data are given by $\boldsymbol{\mu} = [1, 1]^\top$ and $\boldsymbol{\sigma} = [2, 6]^\top$, respectively. The learned parameters are $\mathbf{w}^* = [-6, -9]^\top$ and $w_0^* = 9$.

What is the output of the trained linear regression model for a *new* test point $\mathbf{x}_{test} = [2, 3]^\top$?

- A. 1056
- B. 3
- C. -30
- D. 1023
- E. Don't know.

Model 1: Logistic Regression (LR)

		Predicted	
		Class 1	Class 2
Actual	Class 1	52	64
	Class 2	244	104

Model 2: Decision Tree (DT)

		Predicted	
		Class 1	Class 2
Actual	Class 1	60	56
	Class 2	110	238

Table 3: Confusion matrices.

Question 11. Table 3 shows the confusion matrices related to a Logistic Regression model (LR) and a Decision Tree classifier (DT) trained on a particular instance of the Earthquakes in Romania dataset to predict Earthquake level. The number of observations is $N = 464$. We define $y = 1$ to indicate a *positive outcome* and $y = 2$ to be a *negative outcome*.

Which one of the following statements is *correct*?

- A. Predicting everything as Negative gives worse accuracy than Model 1.
- B. Model 1 has higher recall than Model 2.
- C. Model 2 has higher precision than Model 1.
- D. Model 1 has lower error rate than Model 2.
- E. Don't know.

Question 12. We consider a synthetic 1-dimensional dataset with the following five points:

$$X = \{-1, 5, 8, 12, 16\}$$

Suppose you want to perform k -means clustering with $K = 2$ centroids.

The centroids are initialized at $\mu_1 = 6$ and $\mu_2 = 11$. What will be the total cost, E , (i.e. the sum of squared distances from the observations to their nearest centroid) after the second iteration of k -means?

- A. 12
- B. 50
- C. 58
- D. 70
- E. Don't know.

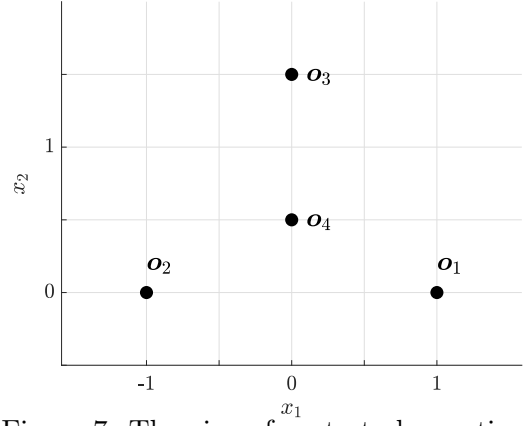


Figure 7: The given four test observations.

Question 13. A logistic regression model was trained on a two dimensional dataset using nonlinear transformations of the two attributes x_1 and x_2 such that for each observation $\mathbf{o} = [x_1, x_2]^\top$ the $p(y = 1 | \mathbf{o}) = \sigma([1, x_1^2, x_2^2]^\top \mathbf{w})$, where σ is the logistic sigmoid. Consider four test observations presented in Figure 7. We know for the associated weights $\mathbf{w} = [w_0, w_1, w_2]^\top$ that the condition $w_1 = w_2$ holds.

Additionally, the following probabilities are known:

$$\begin{aligned} p(y = 1 | \mathbf{o}_1) &\approx 0.62, & p(y = 1 | \mathbf{o}_2) &\approx 0.62, \\ p(y = 1 | \mathbf{o}_3) &\approx 0.85. \end{aligned}$$

Which one of the following statements is *correct*?

Hint: The problem can be solved without complex computations; consider the shape of the decision boundary.

- A. $p(y = 1 | \mathbf{o}_4) > p(y = 1 | \mathbf{o}_1)$
- B. $p(y = 1 | \mathbf{o}_4) < p(y = 1 | \mathbf{o}_1)$
- C. $p(y = 1 | \mathbf{o}_4) = p(y = 1 | \mathbf{o}_3)$
- D. $p(y = 1 | \mathbf{o}_4) > p(y = 1 | \mathbf{o}_3)$
- E. Don't know.

Question 14. In a market basket problem, we consider an association rule $X \rightarrow Y$. Assume that the support of $X \cup Y$ is $\frac{3}{10}$ and the confidence of $X \rightarrow Y$ is $\frac{4}{5}$. What is the support of X ?

- A. $\text{support}(X) = \frac{2}{10}$
- B. $\text{support}(X) = \frac{1}{2}$
- C. $\text{support}(X) = \frac{4}{10}$
- D. $\text{support}(X) = \frac{3}{8}$
- E. Don't know.

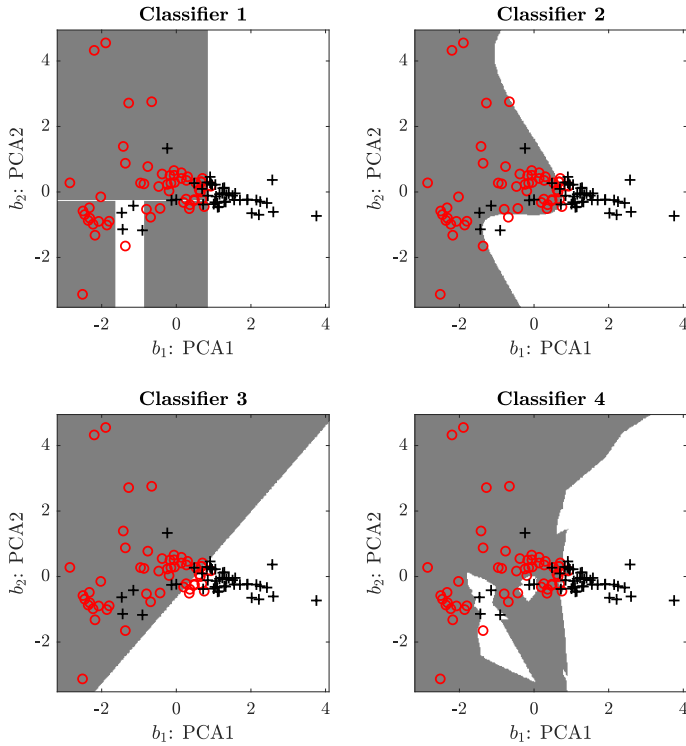


Figure 8: Decision boundaries for four different classifiers trained on the Earthquakes in Romania dataset when projected onto the first two principal components.

Question 15. We will consider a subset of the Earthquakes in Romania dataset (described in Table 1) after it has been projected onto the first two principal components b_1 and b_2 , thereby giving rise to a smaller two-dimensional dataset. We will consider the following four classifiers:

LREG: Logistic regression

ANN: Artificial neural network with 5 hidden units

DT: Decision tree with regular axis-aligned splits ($b_i < c$)

KNN: K-nearest neighbours with $K = 3$

Suppose the classifiers are trained on the two-dimensional dataset and the decision boundary for each of the four classifiers is given in Figure 8. Which one of the following statements is correct?

- A. Classifier 1 corresponds to **DT**,
Classifier 2 corresponds to **KNN**,
Classifier 3 corresponds to **ANN**,

Classifier 4 corresponds to **LREG**.

- B. Classifier 1 corresponds to **DT**,
Classifier 2 corresponds to **LREG**,
Classifier 3 corresponds to **ANN**,
Classifier 4 corresponds to **KNN**.
- C. Classifier 1 corresponds to **DT**,
Classifier 2 corresponds to **ANN**,
Classifier 3 corresponds to **LREG**,
Classifier 4 corresponds to **KNN**.
- D. Classifier 1 corresponds to **KNN**,
Classifier 2 corresponds to **DT**,
Classifier 3 corresponds to **LREG**,
Classifier 4 corresponds to **ANN**.
- E. Don't know.

Question 16. Which one of the following statements about ensemble methods is *correct*?

- A. If the weighted error rate of a classifier in AdaBoost is $\varepsilon_t = 0.5$, then this classifier will have an influence on the result.
- B. In AdaBoost we can prevent overfitting when increasing the iterations if we use a regularized logistic regression classifier with fixed λ as the weak learner.
- C. In bagging and boosting observations are sampled without replacement, but in boosting misclassified observations are given a higher probability of being sampled in the next round.
- D. In AdaBoost, the more errors a classifier makes in a given round of boosting, the less the relative increase of the weights for the misclassified observations.
- E. Don't know.

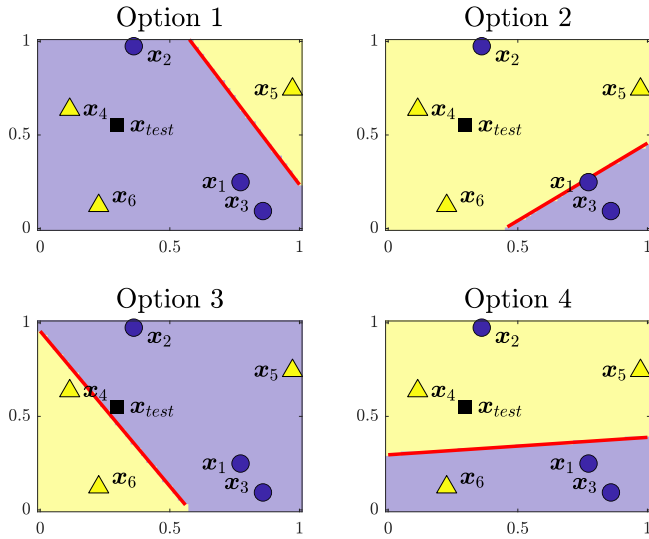


Figure 9: The four possible options for the classifier f_2 . The points of Class 0 are represented by $\bigcirc$ and of Class 1 by $\triangle$. The test point is represented by $\square$.

x	x_1	x_2	x_3	x_4	x_5	x_6
y	0	0	0	1	1	1
w_1	0.166	0.166	0.166	0.166	0.166	0.166
w_2	0.1	0.1	0.1	0.1	0.499	0.1

Table 4: The observations for the AdaBoost model.

Question 17. We consider the dataset in Figure 9 and the weights in Table 4 after two iterations of the AdaBoost algorithm. The predictions for a test point x_{test} for the two weak classifiers are $f_1(x_{test}) = 0$ and $f_2(x_{test}) = 1$. The final prediction using these two weak classifiers is $f^*(x_{test}) = 0$.

Which one of the options in Figure 9 corresponds to the classifier f_2 ?

Hint: Consider the weighted error rates of f_1 and f_2 .

- A. Option 1
- B. Option 2
- C. Option 3
- D. Option 4
- E. Don't know.

Question 18. We consider a subset of $N = 120$ points from the Earthquakes in Romania dataset and two different classifiers (Model 1 and Model 2). We apply two-level cross-validation to estimate the generalization error, with 2 outer folds and 3 inner folds. In Table 5 the

Outer fold $i = 1$				
Inner folds	$j = 1$	$j = 2$	$j = 3$	Test
Model 1	1	9	4	55
Model 2	6	4	19	43

Outer fold $i = 2$				
Inner folds	$j = 1$	$j = 2$	$j = 3$	Test
Model 1	18	4	8	47
Model 2	15	4	9	29

Table 5: The information for the two-level cross-validation. For each outer fold, we are given the number of misclassified points for each model for each inner fold on the associated validation set, together with the number of misclassified points on the associated test partition of the outer fold.

information for each outer fold, model, and inner fold is given, together with the performance of each model on the test partition of the outer fold.

Which one of the following options is a correct estimate of the generalization error?

- A. $\hat{E}^{\text{gen}} = 0.6$
- B. $\hat{E}^{\text{gen}} = 0.7$
- C. $\hat{E}^{\text{gen}} = 0.85$
- D. $\hat{E}^{\text{gen}} = 0.35$
- E. Don't know.

Question 19. Which one of the following statements about cross-validation (CV) is *correct*?

- A. Ideally, the same test set should be used for estimating the parameters of the model and for the final evaluation of the model's performance.
- B. In K -fold cross-validation the trained models are less similar to each other compared to those in leave-one-out cross-validation because the overlap between the training sets is smaller.
- C. Two-fold cross-validation is the same as the hold-out method when 50% is hold-out.
- D. The cross-validation technique is *exclusively* used for supervised learning problems to estimate the generalization error.
- E. Don't know.

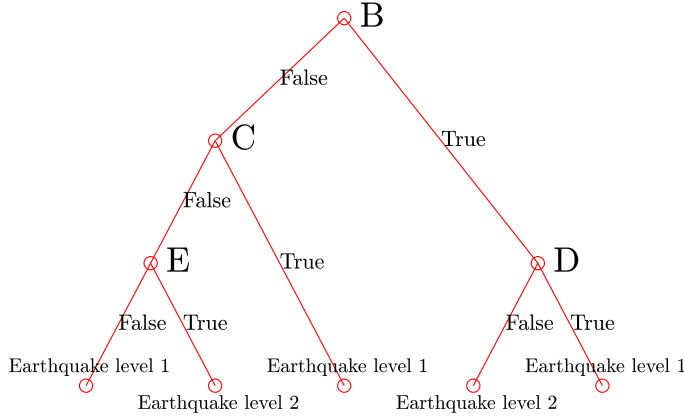


Figure 10: Structure of decision tree. The goal is to determine the splitting rules.

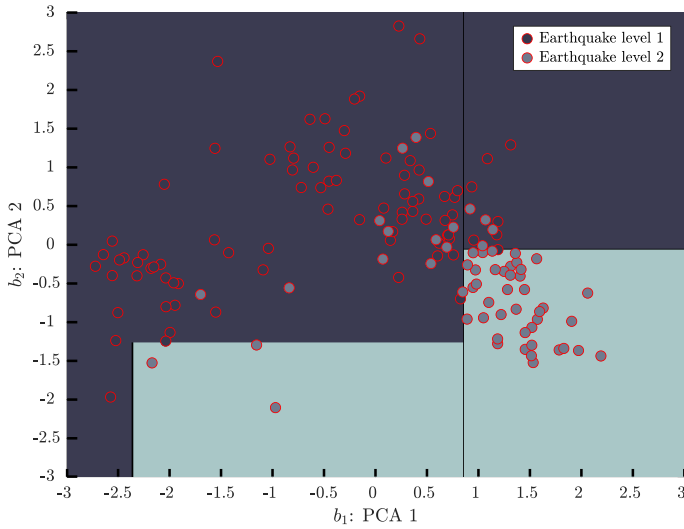


Figure 11: Classification boundary.

Question 20. We consider the Earthquakes in Romania dataset projected onto the first two principal directions. Suppose we train a decision tree to predict which of the two classes an observation belongs to. Since the attributes are continuous, we will consider binary splits of the form $b_i \geq z$ for different values of i and z , where b_1, b_2 refer to the coordinates of the observations when projected onto principal directions. Suppose the trained decision tree has the form shown in Figure 10, and that according to the tree the predicted label assignment for a subset of the observations is as given in Figure 11, what is then the correct rule

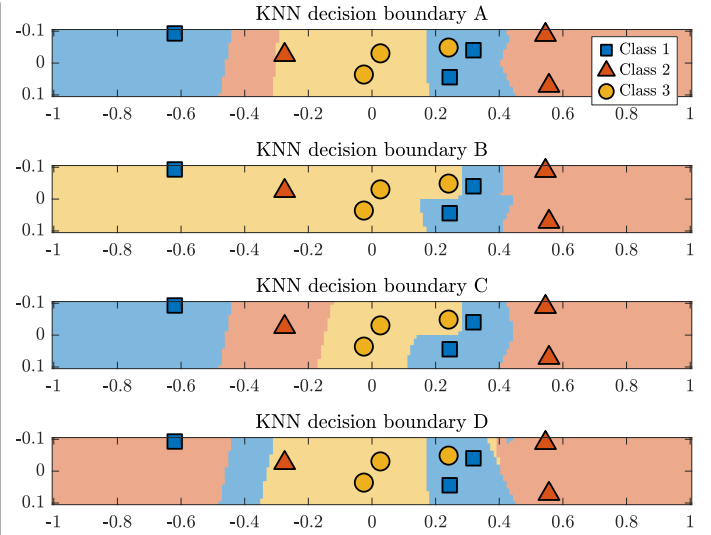


Figure 12: Decision boundaries for four KNN classifiers.

assignment to the nodes in the decision tree?

- A. **B:** $b_2 \geq -1.27$, **C:** $b_2 \geq -0.07$, **D:** $b_1 \geq -2.38$, **E:** $b_1 \geq 0.83$
- B. **B:** $b_1 \geq 0.83$, **C:** $b_2 \geq -0.07$, **D:** $b_2 \geq -1.27$, **E:** $b_1 \geq -2.38$
- C. **B:** $b_1 \geq 0.83$, **C:** $b_2 \geq -1.27$, **D:** $b_2 \geq -0.07$, **E:** $b_1 \geq -2.38$
- D. **B:** $b_2 \geq -0.07$, **C:** $b_1 \geq -2.38$, **D:** $b_2 \geq -1.27$, **E:** $b_1 \geq 0.83$
- E. Don't know.

Question 21. Consider a two-dimensional dataset comprised of $N = 9$ observations shown in Figure 12. The dataset consists of three classes indicated by the blue squares (class 1), red triangles (class 2) and yellow circles (class 3). In the figure, the decision boundaries for four K -nearest neighbor classifiers (KNN) are shown. Which one of the plots correspond to the $K = 3$ nearest-neighbour classifier assuming ties are broken by assigning the point to the *nearest* neighbour's class?

- A. KNN decision boundary A
- B. KNN decision boundary B
- C. KNN decision boundary C
- D. KNN decision boundary D
- E. Don't know.

	o_1	o_2	o_3	o_4	o_5	o_6	o_7	o_8	o_9	o_{10}
o_1	0.0	0.7	1.7	4.0	3.6	3.8	1.2	2.0	2.2	1.5
o_2	0.7	0.0	1.2	3.8	3.4	3.6	1.6	2.4	2.6	1.8
o_3	1.7	1.2	0.0	3.4	2.6	3.2	2.1	2.7	2.9	2.2
o_4	4.0	3.8	3.4	0.0	2.8	0.2	4.5	5.1	5.2	4.7
o_5	3.6	3.4	2.6	2.8	0.0	2.7	4.0	4.6	4.8	4.2
o_6	3.8	3.6	3.2	0.2	2.7	0.0	4.3	4.9	5.0	4.5
o_7	1.2	1.6	2.1	4.5	4.0	4.3	0.0	0.8	1.0	0.4
o_8	2.0	2.4	2.7	5.1	4.6	4.9	0.8	0.0	0.2	0.6
o_9	2.2	2.6	2.9	5.2	4.8	5.0	1.0	0.2	0.0	0.8
o_{10}	1.5	1.8	2.2	4.7	4.2	4.5	0.4	0.6	0.8	0.0

Table 6: The pairwise Euclidian distances, $d(o_i, o_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2 = \sqrt{\sum_{k=1}^M (x_{ik} - x_{jk})^2}$ between 10 observations from the Earthquakes in Romania dataset (recall that $M = 4$). Each observation o_i corresponds to a row of the data matrix $\mathbf{X}$ of Table 1. The colors indicate classes such that the black observations $\{o_1, o_2, o_3, o_4, o_5, o_6\}$ belong to class C_1 (corresponding to Low strength), and the red observations $\{o_7, o_8, o_9, o_{10}\}$ belong to class C_2 (corresponding to High strength). To avoid single features to dominate, the dataset was standardized by subtracting the mean and dividing by the standard deviation.

Question 22. Which one of the following statements about artificial neural networks (ANNs) is *correct*?

- A. In a regression problem, an ANN with linear activation functions ($h^{(i)}(x) = x$) results in a complex model that tends to overfit if output depends on the input in a non-linear manner.
- B. Randomly initializing two ANNs with identical activation functions and number of hidden units, and training them with the same data, will always converge to the same model.
- C. The advantage of ANNs is that they require less data for training and are more interpretable and explainable compared to related techniques, such as linear regression.
- D. Increasing the number of hidden units in an ANN can lead to overfitting if not properly controlled through regularization.
- E. Don't know.

Question 23. Consider the dendrogram from Figure 13. Suppose we apply a cutoff (indicated by the black line in Figure 13) thereby generating three clusters. We wish to compare the quality of this clustering,

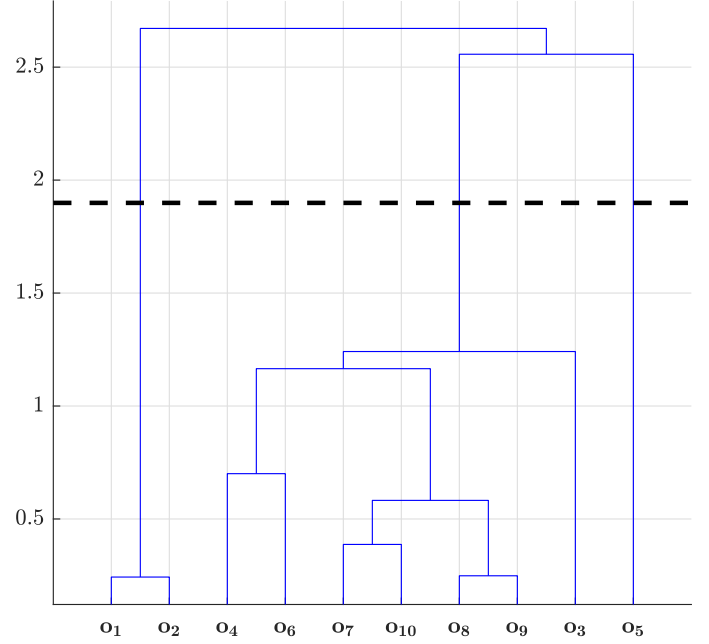


Figure 13: A hierarchical clustering is applied to the 10 observations in Table 6 (using unrounded values) with a cutoff indicated by the dotted line, thereby generating 3 clusters.

Q , to the ground-truth clustering, Z , indicated by the colors in Table 6.

Recall, $\mathbf{n}$ is the count matrix where element n_{km} indicates the number of observations assigned to cluster k in Z and m in Q . S is the number of times Z and Q agree that two observations are in the same cluster. D the number of times Z and Q agree that two observations are not in the same cluster.

Determine which one of the following set of intermediate calculations is correct.

A. $\mathbf{n} = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 4 & 0 \end{bmatrix}$, $S = 10$, $D = 12$

B. $\mathbf{n} = \begin{bmatrix} 0 & 4 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 3 \end{bmatrix}$, $S = 9$, $D = 22$

C. $\mathbf{n} = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, $S = 10$, $D = 16$

D. $\mathbf{n} = \begin{bmatrix} 2 & 4 \\ 1 & 0 \\ 3 & 0 \end{bmatrix}$, $S = 10$, $D = 16$

E. Don't know.

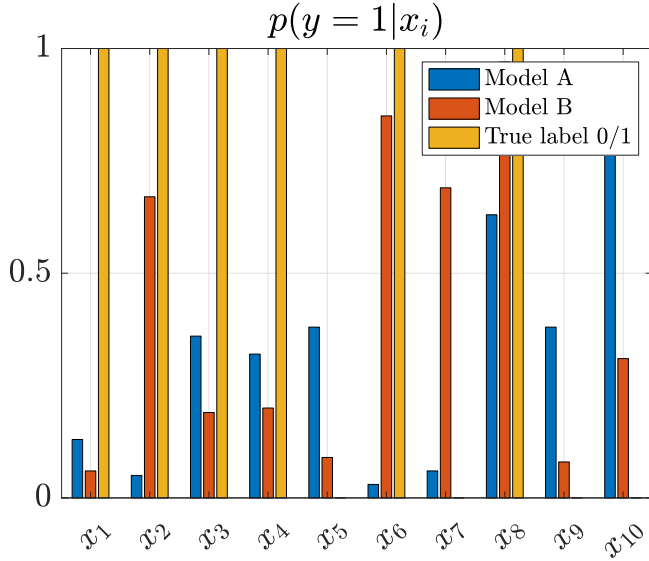


Figure 14: The $p(y = 1|x_i)$ for the two models on a test set of 10 points $x_1, \dots, x_{10}$, and the true test labels. If $p(y = 1|x_i) > 0.5$, then x_i is classified to Class 1.

Question 24. We consider a statistical procedure to determine if there is a significant difference between two classifiers, Model A and Model B, using McNemar's test. The probabilities $p(y = 1 | x_i)$ for the two models on a test set of 10 points are shown in Figure 14, together with the true test label. Our null hypothesis is that the classifiers perform similarly.

Determine which combination of estimated difference in accuracy $\hat{\theta}$, confidence interval (CI) with $\alpha = 0.05$, and p-value is the only plausible answer.

- A. $\hat{\theta} = 0.4$, $CI = [-0.33, 1.0]$, p-value < 0.63
- B. $\hat{\theta} = -0.2$, $CI = [-0.54, 0.16]$, p-value > 0.5
- C. $\hat{\theta} = 0.36$, $CI = [-0.54, 0.16]$, p-value < 0.63
- D. $\hat{\theta} = -0.2$, $CI = [-0.54, 0.16]$, p-value < 0.01
- E. Don't know.

Question 25. A neural network classifier is trained to distinguish between two classes $y \in \{1, 2\}$ and produce class-probability $\hat{y}$ and the receiver operator characteristic (ROC) curve of the network when evaluated on a test set with $N = 8$ observations is shown in Figure 16. Suppose we plot the predictions on the $N = 8$ test observations by their $\hat{y}$ value along the x -axis and indicate the class labels by either a black circle (negative class $y = 1$) or red cross (positive class $y = 2$), which one of

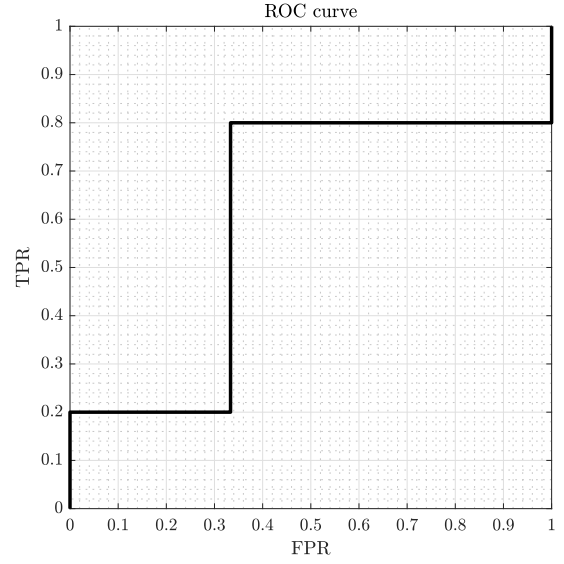


Figure 15: ROC curve for a neural network classifier, where the predictions and true class labels are one of the options in Figure 16.

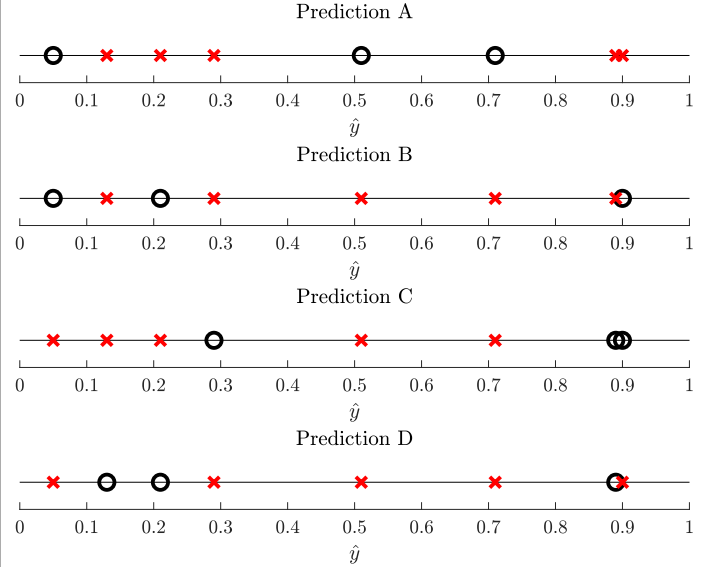


Figure 16: Four candidate predictions for the ROC curve in Figure 15. The observations are plotted horizontally, such that the position on the x -axis indicate the predicted probability $\hat{y} = p(y = 2|x)$, and the marker/color indicate the class membership, such that the black circles indicate the observation belongs to the negative class $y_i = 1$ and red crosses to the positive class $y_i = 2$.

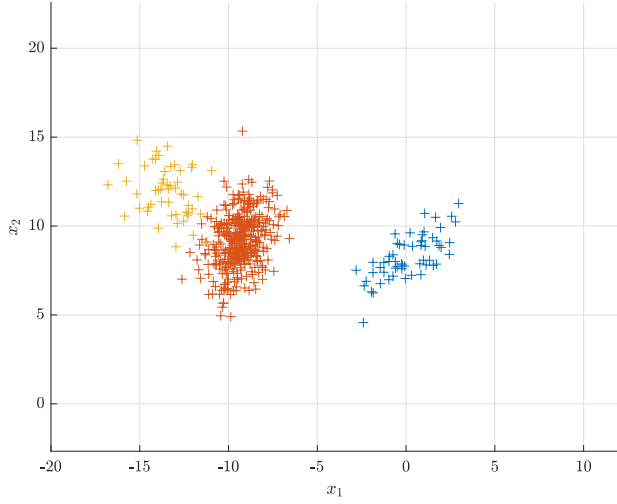


Figure 17: 500 observations drawn from a Gaussian Mixture Model (GMM) with three clusters.

the subplots in Figure 16 then corresponds to the ROC curve in Figure 15?

- A. Prediction A
- B. Prediction B
- C. Prediction C
- D. Prediction D
- E. Don't know.

Question 26. We consider an artificial neural network with the following first layer weights $\mathbf{w}_1^{(1)} = [-3, -1]^\top$ and $b_1^{(1)} = 3$ and $\mathbf{w}_2^{(1)} = [1, 0]^\top$ and $b_2^{(1)} = 2$, and the second layer $\mathbf{w}^{(2)} = [-2, -2]^\top$ and $b^{(2)} = 1$. We consider the ReLU activation $h^{(1)}(x) = \max\{0, x\}$ in the first layer and the identity activation in the second layer $h^{(2)}(x) = x$, and a test point $\mathbf{x}_{test} = [3, 3]^\top$. What is the correct output of the neural network?

- A. $f(\mathbf{x}_{test}; \mathbf{w}) = -9$
- B. $f(\mathbf{x}_{test}; \mathbf{w}) = 9$
- C. $f(\mathbf{x}_{test}; \mathbf{w}) = 1$
- D. $f(\mathbf{x}_{test}; \mathbf{w}) = -27$
- E. Don't know.

Question 27. Let $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denote the multivariate normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. In Figure 17 is given 500 observations drawn

from a density defined by a Gaussian Mixture Model (GMM) with three clusters. Each observation is colored in terms of which cluster it came from in the Gaussian Mixture model.

Which one of the following GMM densities was used to generate the data?

A.

$$p(\mathbf{x}) = \frac{1}{10}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 0.2 \\ 8.2 \end{bmatrix}, \begin{bmatrix} 1.4 & -0.4 \\ -0.4 & 2.0 \end{bmatrix}\right) + \frac{4}{5}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -9.3 \\ 9.3 \end{bmatrix}, \begin{bmatrix} 2.4 & 1.0 \\ 1.0 & 1.4 \end{bmatrix}\right) + \frac{1}{10}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -13.3 \\ 11.8 \end{bmatrix}, \begin{bmatrix} 1.0 & 0.4 \\ 0.4 & 2.5 \end{bmatrix}\right)$$

B.

$$p(\mathbf{x}) = \frac{4}{5}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 0.2 \\ 8.2 \end{bmatrix}, \begin{bmatrix} 1.0 & 0.4 \\ 0.4 & 2.5 \end{bmatrix}\right) + \frac{1}{10}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -9.3 \\ 9.3 \end{bmatrix}, \begin{bmatrix} 2.4 & 1.0 \\ 1.0 & 1.4 \end{bmatrix}\right) + \frac{1}{10}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -13.3 \\ 11.8 \end{bmatrix}, \begin{bmatrix} 1.4 & -0.4 \\ -0.4 & 2.0 \end{bmatrix}\right)$$

C.

$$p(\mathbf{x}) = \frac{1}{10}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 0.2 \\ 8.2 \end{bmatrix}, \begin{bmatrix} 2.4 & 1.0 \\ 1.0 & 1.4 \end{bmatrix}\right) + \frac{4}{5}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -9.3 \\ 9.3 \end{bmatrix}, \begin{bmatrix} 1.0 & 0.4 \\ 0.4 & 2.5 \end{bmatrix}\right) + \frac{1}{10}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -13.3 \\ 11.8 \end{bmatrix}, \begin{bmatrix} 1.4 & -0.4 \\ -0.4 & 2.0 \end{bmatrix}\right)$$

D.

$$p(\mathbf{x}) = \frac{1}{10}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 0.2 \\ 8.2 \end{bmatrix}, \begin{bmatrix} 1.0 & 0.4 \\ 0.4 & 2.5 \end{bmatrix}\right) + \frac{1}{10}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -9.3 \\ 9.3 \end{bmatrix}, \begin{bmatrix} 1.4 & -0.4 \\ -0.4 & 2.0 \end{bmatrix}\right) + \frac{4}{5}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -13.3 \\ 11.8 \end{bmatrix}, \begin{bmatrix} 2.4 & 1.0 \\ 1.0 & 1.4 \end{bmatrix}\right)$$

E. Don't know.